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TOS DESIGN OPTIMIZATION: PLANT DIVERSITY SAMPLING

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1 BACKGROUND AND GOALS

The National Ecological Observatory Network (NEON) generates data and samples to advance understanding of ecological processes and inform management of US ecosystems. Multiple tiers of requirements provide traceable links between this high-level NEON mission statement and the protocols that direct data collection. The Terrestrial Observation System (TOS) science design for plant diversity ([RD03]) and the resulting study design, protocol ([RD03]), and data product (Plant Presence and Percent Cover, DP1.10058.001) are designed to enable status and trends in composition, abundance, and richness of plant species and function types at multiple scales.

1.1 Description of sample design and available data

NEON's plot-based sampling of plant species diversity satisfies the requirements that "NEON's measurement strategy will include biological invaders and diversity of key organismal groups" and "NEON measurements of plant diversity will include native and invasive plant species (Schimel et al. 2011)." Species are observed with the Carolina Vegetation Survey (Peet et al. 1998) multi-scale plot design that borrows from a sampling approach pioneered by Robert Whittaker (Shmida 1984, Stohlgren 1995). NEON samples with a 20 x 20m square plot with four 10m x 10m subplots containing nested 10m² and 1m² subplots (Figure 1). Specifically:

- The identity of each species (or the lowest taxonomic rank possible) according to naming conventions maintained by the US Department of Agriculture, Natural Resources Conservation Service PLANTS database (USDA, NRCS 2016) is recorded in each subplot - eight 1m², eight 10m², four 100m².
- Estimates of taxon-specific abundance is made with estimates of cover within the 1m² subplots.

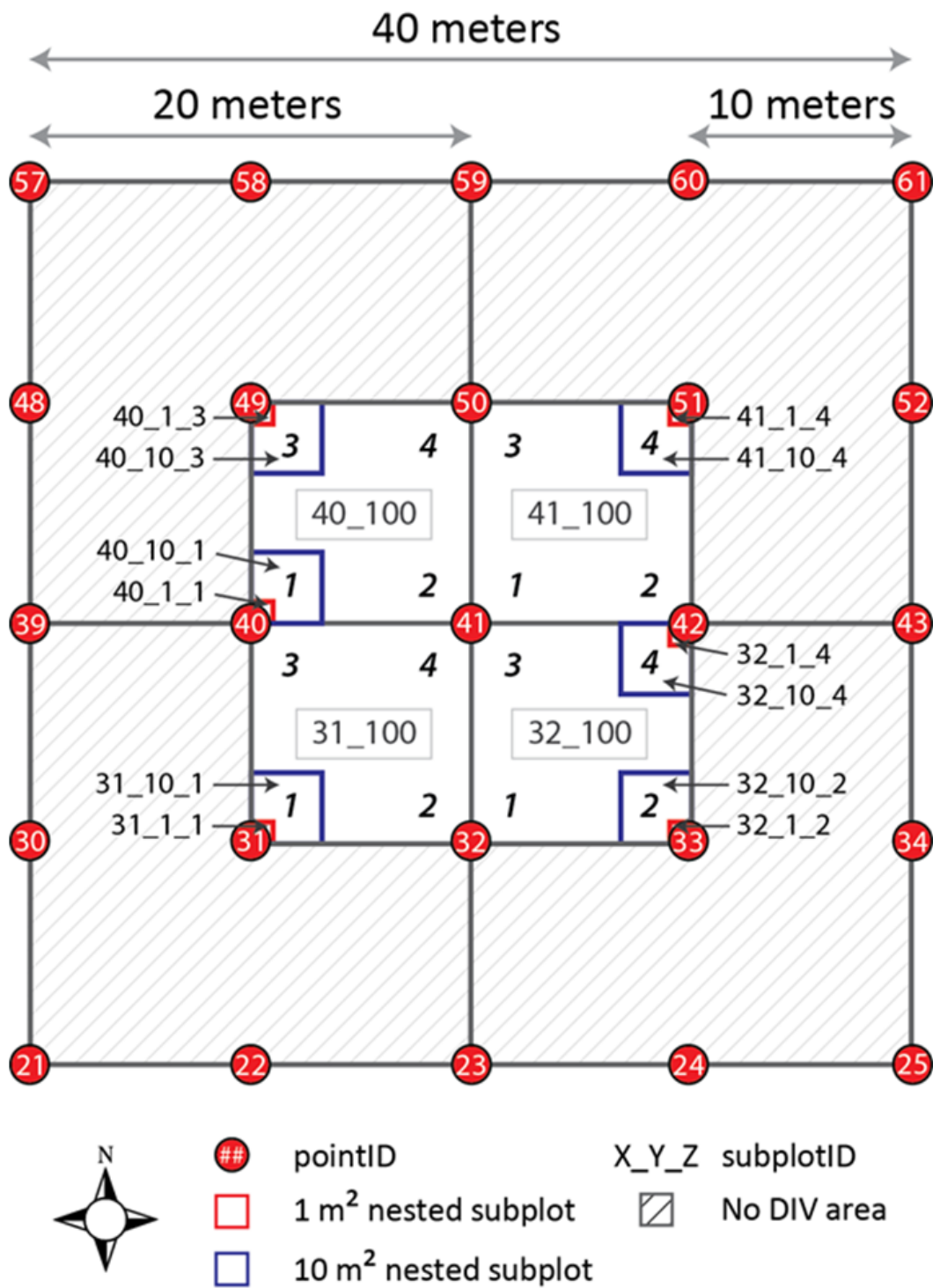


Figure 1: A diagram of the multiscale plot for observations of plant cover-abundance and occurrence data.

At each of 47 terrestrial NEON sites, the NEON spatial sampling design for organisms and soil directs observations to (Thorpe et al. 2016):

- Tower Base Plots: Randomly placed plots near the NEON flux tower designed to calibrate sensor-based measures of atmosphere and soil
- Distributed Base Plots: Plots to measure variability across larger landscapes according to a stratified-random design

The plant diversity protocol is implemented at three of the 20–30 Tower Base Plots. The majority of the plant species diversity sampling effort occurs at the Distributed Base Plots (Thorpe et al. 2016). Ecological literature (e.g., Stohlgren 2007), early input from the ecological community, and power analyses (Thompson et al. 2012, Barnett et al. 2018) resulted in an initial sample size of 20–30 Distributed Base plots at each site.

The frequency and timing of sampling is guided by NEON requirements (Schimel et al. 2011). Implementing the protocol at each plot at least one time each year enables the quantification of annual rates of change. These sampling bouts occur during a one to two month period when the majority of species flower or possess other parts conducive to identification. This sampling window targets annual peaks in greenness as measured by MODIS, and incorporates input from local ecologists and NEON phenology observations (Meier et al. 2023). The few NEON sites characterized by distinct bimodal patterns in peak greenness populated disparate suites of apparent species are sampled twice annually. For example, the Sonoran desert (represented by the NEON site at the Santa Rita Experimental Range) gets rain in late winter and a monsoonal moisture pulse late in the summer. Plant species adapted to the particular intersection of temperature and the timing and amount of precipitation grow and flower and can be identified only during one of the two distinct events (Ogle and Reynolds 2004).

For the purpose of these analyses, data includes all available Distributed Base Plots from all 47 NEON sites (NEON 2025) downloaded directly from the application programming interface (API) and stacked with 'neonUtilities::stackByTable()' function on 03/01/2026.

1.2 Analytical Goals

Statistically rigorous analyses are needed to assess the capacity of NEON data to address Observatory goals and to guide iterative sampling design optimization efforts. Initial spatial and temporal sampling designs for the NEON Terrestrial Observation System (TOS) and Aquatic Observation System (AOS) were developed in collaboration with Technical Working Groups (TWGs) comprised of community experts, and were captured in Science Design documents (RD[01]). The initial designs relied on analysis of published datasets (where relevant), analysis of NEON prototype data collection efforts, and subject matter expertise.

As the NEON Observatory matures, it is critical for initial design assumptions to be tested with multiple years of data collected from NEON sites. Analyses must:

- Describe where the data support reduction of sampling effort without compromising NEON's ability to meet design requirements (e.g., target sites for reduction in sampling effort, spatial scales where replication is not necessary, etc.).

- Evaluate effects of design modifications on uncertainty and ability to detect changes for key response variables (e.g., effects of reduced spatial and/or temporal sampling effort).

Analysis and evaluation of the data provides a feedback loop that enables assessment of the Designs relative to Observatory goals. Moreover, results of these analyses will allow the NEON to effectively prioritize sampling costs in the face of uncertain future funding levels.

Specification of analyses suitable for evaluating the efficacy of the NEON sampling designs required development of specific questions. Power analyses and sampling simulations were then used to determine the degree to which design modifications affected the ability of the data to address those questions. Questions and analytical approaches were developed in consultation with NEON Technical Working Groups comprised of subject matter experts selected from the ecological community.

The two primary and actionable questions of a 2018 effort for these data were:

1. Where is the variability in the data? Specifically, what proportion of the variance at each site exists at the year, bout, plot, and subplot scales? Understanding the components of variance provides a critical component of the actionable directions for modifying sampling size, subplot sampling intensity, and temporal sampling frequency.
2. Is it possible to detect meaningful interannual variation in key response variables? Understanding the power to detect change between years provides a direct indication of the sample sizes required to detect meaningful ecological change.

The variance components analysis supported the reduction in the number of the number nested 10m² and 1m² subplots within each plot from eight to six. Additionally, the 10m² and 100m² subplots at which species presence are recorded were reduced to every other year sampling. The 1m² subplots at each plot continue to be sampled annually.

The power analysis reduced the total number of plots sampled at a small number of sites. Only sites with three or more years of data were considered. An initial goal was to evaluate trend detection, but it was decided insufficient years of data had been collected for that analysis.

With many more years of data available across all NEON terrestrial sites in 2026 (Appendix A), this effort revisits the question of detectability of interannual change in plant community composition and abundance. Specifically, we ask:

1. What level of spatial replication (number of plots) is required to reliably detect a specified relative change in plant species cover at a site?
2. What level of spatial replication is required for reliable estimation of multi-year directional trends in species cover?

Together, these analyses provide complementary bounds on the minimum sampling effort required to detect meaningful ecological change at short (interannual) and long (multi-year trend) temporal horizons.

2 RELATED DOCUMENTS AND ACRONYMS

2.1 Reference Documents

RD[01]	NEON.DOC.000001	NEON Observatory Design
RD[02]	NEON.DOC.002652	NEON Level 1, Level 2, and Level 3 Data Products Catalog
RD[03]	NEON.DOC.000912	TOS Science Design for Plant Diversity
RD[04]	NEON.DOC.014042	TOS Protocol and Procedure: Plant Diversity Sampling

2.2 Acronyms

Acronym	Definition
NEON	National Ecological Observatory Network
TOS	Terrestrial Observation System
TWG	Technical Working Group
NLCD	National Land Cover Database
GJAM	Generalized Joint Attribute Model
CA	Censored Abundance (GJAM likelihood for cover data)
MCMC	Markov Chain Monte Carlo

3 METHODS

We evaluated the capacity of NEON’s terrestrial plant sampling design to detect ecological change in species- and community-level response variables using a hierarchical Bayesian modeling framework combined with simulation-based power and sensitivity analyses. The assessment comprises two complementary analyses:

1. We conducted a perturbation-based power analysis to quantify the spatial replication (number of randomly placed plots) required to detect a specified magnitude of interannual change in species cover, evaluated through a paired counterfactual design that controls for natural year-to-year variability.
2. We evaluated the spatial replication required for reliable estimation of multi-year directional trends, including the stability of trend estimates across sample sizes and the detectability of imposed trends of known magnitude.

Together, these analyses address the two primary temporal scales at which change detection is relevant to NEON’s mission: abrupt interannual shifts and gradual directional trends.

3.1 Data and preprocessing

The observation unit for modeling is a species-specific cover value at the plot-year level. Within each plot-year, the six (or eight prior to 2019) 1m² subplot observations are averaged to produce a single cover value per species per plot per year. For example, at a site where 30 plots are sampled annually for 8

years, a species recorded on all plots would contribute 240 plot-year observations to the model. The full model input for a site is the matrix of all species $\times$ all plot-years, with cover values as entries.

Prior to model fitting, species occurring in fewer than three plots across the entire site record are aggregated into a composite “OTHER” category. This filter removes species for which spatial replication is insufficient to estimate covariance structure, while retaining them in the modeled community so that their aggregate contribution to total cover is represented (Clark et al. 2017). Species with all-zero cover or zero variance after aggregation are removed. The “OTHER” category participates in model fitting but is excluded from power and trend analyses, which focus on individually identifiable species.

Separately, a cover floor of 3% mean raw field cover is applied when evaluating power and trend stability. Species below this threshold are excluded from the power analysis because a 20% relative change at 3% cover corresponds to 0.6 percentage points—below the 1% recording resolution of NEON’s field protocol. This cover floor is computed from the raw field observations (mean of percent cover per species across all plot-years at the site) and is independent of the GJAM model predictions which undergo compositional normalization that compresses cover values.

3.2 Joint species distribution model

Both analyses share a common modeling framework. Plant species cover was modeled with generalized joint attribute models (GJAM; Clark et al. 2017), a multivariate hierarchical Bayesian framework that jointly represents all species at a site while accounting for shared responses to environmental covariates and residual covariance among species. A single GJAM model is fit per site using all available plot-years. Both the perturbation-based and trend analyses draw on the posterior output of this shared fit. For plot p in year t , the vector of observed percent cover values for all S species is modeled as:

$$\mathbf{y}_{p,t} \sim \text{CA}(\mu_{p,t}, \Sigma)$$

$$\mu_{p,t} = \mathbf{X}_{p,t} \mathbf{B}$$

where $\mathbf{y}_{p,t}$ is the S -vector of observed percent cover, $\mu_{p,t}$ is the latent mean cover vector, $\mathbf{X}_{p,t}$ is the design matrix of covariates, $\mathbf{B}$ is the $K \times S$ matrix of species-specific regression coefficients (equivalent to the transpose of the coefficient matrix in Clark et al. 2017), and Σ represents residual covariance among species not explained by the covariates, where K is the number of covariates.

The censored-abundance (CA) likelihood represents observed cover as a censored realization of an underlying multivariate normal latent variable, allowing exact zeros as well as continuous positive values.

Two model specifications were used depending on land cover heterogeneity at each site: $\mathbf{Y} \sim \text{year} + \text{nlcdClass}$ at sites with multiple National Land Cover Database (NLCD) classes represented among sample plots, and $\mathbf{Y} \sim \text{year}$ at sites with a single class or where the multi-class specification failed to converge. Year was treated as a categorical factor. The MCMC sampler was run for 5,000 iterations; 2,500 iterations were discarded as burn-in, yielding 2,500 retained posterior draws. For downstream posterior prediction and sensitivity analyses, we used $G = 1,000$ posterior draws sampled from the retained MCMC output for computational efficiency.

3.2.1 Posterior prediction and dimension reduction

Joint modeling of plant communities at NEON sites frequently involves hundreds of species observed across tens of plots (Appendix here to numbers of sp), producing covariance matrices that are high-dimensional relative to the available sample size. To illustrate, consider a southeastern forest site with 400 plant species observed across 30 plots per year: estimating the 400×400 residual covariance matrix Σ from 300 plot-year observations is not feasible using standard approaches.

GJAM addresses this challenge through a Bayesian dimension reduction framework (Taylor-Rodriguez et al. 2017; Clark et al. 2017). Rather than estimating Σ directly, the model represents the S -dimensional species response through a lower-dimensional factor structure. Species are assigned to N_c clusters via a truncated Dirichlet process representation, and within each cluster, an r -dimensional factor loading matrix $\mathbf{Z}$ captures shared covariance. The implied covariance is:

$$\Sigma = \mathbf{A}\mathbf{A}' + \sigma_\varepsilon^2 \mathbf{I}$$

where $\mathbf{A} = \mathbf{Q}(\mathbf{k})\mathbf{Z}$ is the $S \times r$ loading matrix that maps the reduced factor structure back to the full species space via cluster assignments $\mathbf{k}$, and σ_ε^2 is a scalar residual variance. This factorization reduces the effective number of covariance parameters from $S(S+1)/2$ to $(N_c \times r) + 1$, enabling stable estimation even when S greatly exceeds the number of observations n .

GJAM invokes dimension reduction when the number of species is large relative to the number of observations, which was the case for all NEON sites analyzed here (Taylor-Rodriguez et al. 2017). This condition was met at all 46 NEON sites, where species counts ranged from 39 to 406 while plot-year observations ranged from approximately 100 to 400. We therefore set reduction explicitly with a truncation level of $N_c = 20$ clusters and $r = 5$ factors across all sites, ensuring consistent parameterization and comparability of posterior predictions across the Observatory. Posterior predictive draws were reconstructed from the stored MCMC chains for $\mathbf{B}$, $\mathbf{k}$, $\mathbf{Z}$, and σ_ε^2 using the reduced-rank generative model:

$$\mathbf{y}_i^{(g)} = \mathbf{x}_i \mathbf{B}^{(g)} + \mathbf{w}_i^{(g)} \mathbf{A}^{(g)'} + \varepsilon_i^{(g)}$$

where $\mathbf{x}_i \mathbf{B}^{(g)} = \mu_i^{(g)}$ is the latent mean response defined above, $\mathbf{B}^{(g)}$ is the $K \times S$ matrix of species-specific regression coefficients for posterior draw g , $\mathbf{A}^{(g)}$ is the $S \times r$ loading matrix reconstructed from cluster assignments $\mathbf{k}^{(g)}$ and factor loadings $\mathbf{Z}^{(g)}$ at draw g , $\mathbf{w}_i^{(g)}$ is an r -dimensional latent factor score with $\mathbf{w}_i^{(g)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$, and $\varepsilon_i^{(g)}$ is species-level residual error with $\varepsilon_i^{(g)} \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^{2(g)} \mathbf{I})$. Superscript (g) indexes the posterior draw used for prediction ($g = 1, \dots, G$, with $G = 1,000$ in the sensitivity analyses). For each draw, the parameters $\mathbf{B}^{(g)}$, $\mathbf{A}^{(g)}$, and $\sigma_\varepsilon^{2(g)}$ are fixed, while $\mathbf{w}_i^{(g)}$ and $\varepsilon_i^{(g)}$ are drawn independently, yielding posterior predictive realizations that preserve the covariance structure implied by the reduced-rank model without explicitly forming the full $S \times S$ covariance matrix.

The analysis propagates key sources of uncertainty through the posterior predictive distribution. Parameter uncertainty is captured through variation across MCMC draws, while residual variability—representing ecological and measurement variation not explained by covariates—is captured through the stochastic

components $\mathbf{w}$ and ε . Observation uncertainty (field measurement error) is not modeled explicitly but is implicitly absorbed into the residual variance estimates.

3.3 Analytical framework

Both analyses use a common modeling and prediction framework to evaluate the detectability of ecological change. Posterior predictive draws from a single site-level model provide a representation of expected species abundance under observed environmental and spatial variability. These predictions are then combined with controlled perturbations and plot-level subsampling to quantify how sampling intensity influences inference.

The two analyses differ only in the temporal structure of the signal being evaluated. The counterfactual analysis focuses on discrete changes between paired states, while the trend analysis evaluates directional change accumulated over multiple years. In both cases, inference is based on the behavior of posterior predictive summaries under repeated subsampling of plots, allowing direct comparison of how sampling design affects the ability to detect and estimate change.

3.4 Year-to-year change detection (counterfactual analysis)

The objective of the counterfactual analysis is to evaluate the extent to which changes in species abundance can be detected given the spatial variability observed within NEON sites. Because plant communities are spatially heterogeneous, the detectability of change depends not only on the magnitude of the signal, but also on the number and configuration of plots used to estimate site-level means.

To address this, we used posterior predictive draws from a single site-level model to generate paired baseline and perturbed realizations of species cover. The perturbation represents a controlled, biologically meaningful change applied to the latent mean structure, allowing direct comparison of predicted site-level responses under identical underlying variability.

For each site, detection was evaluated as a function of the number of plots sampled per year (K). Subsampling plots introduces variability analogous to field sampling, allowing the analysis to quantify how sampling intensity influences the probability of detecting a specified change. The result is a set of detection curves that describe the relationship between sampling effort and the probability of detecting a change of magnitude δ .

This framework isolates the role of spatial sampling in determining detectability. Because the baseline and perturbed predictions are derived from the same posterior draws, differences between them reflect only the imposed change and the variability introduced by subsampling, rather than differences in model fit or parameter uncertainty.

This analysis was motivated by earlier approaches that evaluate change between consecutive years, but extends that framework by removing the confounding influence of interannual ecological variability. Rather than comparing observations from different years, both baseline and perturbed predictions are generated from the same underlying data and posterior parameter draws, with independent realizations of observation-level variability ($\mathbf{w}$ and ε). This counterfactual design isolates statistical detectability from ecological variability among years, providing a controlled evaluation of monitoring performance under realistic covariance structure.

Let $\mu_{s,p}^{(g)}$ denote the posterior predictive cover for species s at plot p for MCMC draw g .

The perturbation was applied directly to the posterior predictive draws as a per-species additive shift proportional to each species' mean raw field cover:

$$\mu_{s,p}^{\text{pert},(g)} = \mu_{s,p}^{(g)} + \delta \cdot \bar{c}_s$$

where $\mu_{s,p}^{(g)}$ is the posterior predictive cover for species s at plot p for draw g , $\mu_{s,p}^{\text{pert},(g)}$ is the corresponding perturbed value, $\bar{c}_s$ is the mean percent cover of species s computed from raw field observations across all plot-years at the site, and δ is the relative effect size. A species at 10% mean cover with $\delta = 0.20$ receives a +2 percentage point perturbation; a species at 60% mean cover receives +12 percentage points. This per-species scaling ensures every species faces the same proportional challenge regardless of baseline cover level. We tested effect sizes of $\delta \in \{0.15, 0.20, 0.25, 0.30\}$, corresponding to 15–30% relative changes in cover.

For each site and year, the baseline and perturbed prediction sets were subsampled to represent alternative levels of spatial replication. All plots sampled in a given year were eligible for subsampling; no requirement was imposed that plots be sampled in consecutive years, as the paired counterfactual design generates both prediction sets from the same year's data. Subsamples consisted of K plots drawn without replacement, with $K \in \{30, 25, 20, 15, 10, 8, 6, 5\}$, and each subsampling was repeated 50 times.

For each Monte Carlo replicate, the site-level mean cover for species s was computed for both the baseline and perturbed prediction sets by averaging across the K sampled plots within each posterior draw g :

$$\bar{\mu}_{s,K}^{(g)} = \frac{1}{K} \sum_{p=1}^K \mu_{s,p}^{(g)}, \quad \bar{\mu}_{s,K}^{\text{pert},(g)} = \frac{1}{K} \sum_{p=1}^K \mu_{s,p}^{\text{pert},(g)}$$

where $\mu_{s,p}^{(g)}$ and $\mu_{s,p}^{\text{pert},(g)}$ denote the baseline and perturbed posterior predictive cover values, respectively.

Detection was evaluated using a per-species absolute difference criterion. For each species s , detection probability was defined as:

$$P\left(\left|\bar{\mu}_{s,K}^{\text{pert},(g)} - \bar{\mu}_{s,K}^{\text{base},(g)}\right| \geq \tau_s\right) \geq 0.80$$

where $\bar{\mu}_{s,K}^{\text{base},(g)}$ and $\bar{\mu}_{s,K}^{\text{pert},(g)}$ are the site-level mean cover values computed from the baseline and perturbed prediction sets, respectively, for posterior draw g , and τ_s is the species-specific detection threshold. The probability is computed as the fraction of posterior draws g for which the absolute difference exceeds τ_s . The ratio of imposed effect to detection threshold is constant across species (2:1 at $\delta = 0.20$), providing equivalent statistical difficulty regardless of cover level. This ensures that detectability comparisons are not driven by differences in species abundance scale.

A species was classified as detectable at sample size K and effect δ when the above criterion was satisfied in at least 80% of the 50 Monte Carlo replicates (power ≥ 0.80). The community-level metric reports the percentage of species (above the 3% cover floor) that are individually detectable at each K .

3.5 Multi-year trend analysis

3.5.1 Objective and conceptual framing

Where the counterfactual analysis evaluates the detectability of discrete changes, the trend analysis addresses a complementary question: whether directional change in species abundance can be reliably estimated over time. In this context, the focus shifts from detecting a difference between two states to estimating the rate of change across multiple years.

As with the counterfactual analysis, inference is based on posterior predictive draws from a single site-level model. For each species, predicted cover values are generated for all plot–year combinations, preserving the spatial and temporal structure of the observed data. These predictions are then used to estimate annual site-level means under varying levels of spatial replication.

To evaluate the influence of sampling intensity, plots are subsampled independently within each year, and site-level mean cover is computed for each posterior draw. A linear trend is then estimated across years for each species, yielding a posterior distribution of slope parameters that reflects both model uncertainty and the variability introduced by spatial subsampling.

Two complementary properties of the estimated trend are evaluated. First, detection assesses whether the magnitude of the estimated slope exceeds a specified threshold, analogous to the threshold-based detection used in the counterfactual analysis. Second, stability evaluates the sensitivity of the estimated slope to the selection of plots, quantified as the coefficient of variation of slope estimates across repeated subsamples at each sample size.

Together, these metrics describe both the ability to detect directional change and the reliability of the estimated rate of change. As in the counterfactual analysis, the objective is not to optimize model fit, but to quantify how spatial sampling constrains inference about ecological change.

Many ecological changes of interest to NEON manifest as gradual directional trends that accumulate over multiple years. A site may experience a slow decline in native grass cover or a progressive increase in woody encroachment that is not detectable in any single year-to-year comparison but becomes apparent over a longer observation window. This analysis addresses the question of what level of spatial replication is required for reliable estimation and detection of such trends.

This question was identified as a priority during the 2018 NEON design optimization effort (Barnett et al. 2018) but could not be addressed at that time due to limited temporal replication. With 8–12 years of data now available at most NEON sites, the question is now tractable.

3.5.2 Moving-window framework

Trend detection was evaluated using a moving-window approach (Urquhart et al. 1998). Rather than fitting a single trend line through the entire time series, we applied overlapping windows of fixed width W

that slide across the available years. This approach reflects realistic monitoring practice: an analyst examining NEON data in any given year would assess the most recent W years to determine whether a trend is present. It also accommodates the possibility that trend strength varies across the observation period.

For each site, we evaluated window widths of $W \in \{5, 10\}$ years. Within each window, a trend was quantified as the ordinary least-squares slope of species cover against year, computed from posterior predictive site-year means derived from plot-level predictions. This yields a posterior distribution of slope estimates for each species, window position, and sample size.

3.5.3 Observed trend detection

The observed trend detectability analysis evaluates whether trends already present in the posterior predictions are detectable as sample size changes. For each species, window position, posterior draw, and sample size K , the site-level mean cover was computed by averaging across the K sampled plots within each year:

$$\bar{\mu}_{s,t,K}^{(g)} = \frac{1}{K} \sum_{p=1}^K \mu_{s,p,t}^{(g)}$$

where $\mu_{s,p,t}^{(g)}$ is the posterior predictive cover for species s at plot p in year t for MCMC draw g . A temporal trend was then estimated for each species and posterior draw as the ordinary least-squares slope of $\bar{\mu}_{s,t,K}^{(g)}$ against centered year:

$$\gamma_{s,K}^{(g)} = \frac{\sum_t (t - \bar{t}) \bar{\mu}_{s,t,K}^{(g)}}{\sum_t (t - \bar{t})^2}$$

Detection was defined as the posterior probability that the absolute slope exceeded a biologically meaningful threshold:

$$P\left(|\gamma_{s,K}^{(g)}| \geq \frac{\delta}{T}\right) \geq 0.80$$

where $\gamma_{s,K}^{(g)}$ is the estimated trend slope for species s at sample size K , $\delta = 0.20$ corresponds to a cumulative 20% change over the window, and $T = t_{\max} - t_{\min}$ is the duration of the observation window.

Plots were subsampled independently within each year, rather than requiring consistent plot identity across time. This accommodates variation in sampling effort among years (including reduced sampling in 2020) and reflects realistic monitoring conditions.

The evaluated sample sizes (K) span the range of sampling effort observed across sites, with a maximum target of $K = 30$. At sites with fewer available plots, the maximum evaluated K corresponds to the number of plots available. Thus, values reported at $K = 30$ represent the highest achievable sampling intensity for each site.

Observed trend detectability answers the question: given the trends already present at a site, how many plots would be needed to detect them with high probability?

This formulation evaluates detectability under the observed temporal structure of the data. A complementary imposed trend analysis, described below, evaluates detectability under a standardized change scenario.

3.5.4 Imposed trend detection

The imposed trend detectability analysis evaluates the ability to detect a trend of known magnitude that may not be present in the observed data. This provides a prospective analysis analogous to the counterfactual change detection analysis, but for gradual directional trends rather than discrete changes at a single time point. For each window, an additive linear trend was imposed on the posterior predictive means before estimating slopes:

$$\mu_{s,p,t}^{\text{trend},(g)} = \mu_{s,p,t}^{(g)} + \frac{\delta}{T}(t - t_0)$$

where $\mu_{s,p,t}^{(g)}$ is the original posterior predictive cover, $\mu_{s,p,t}^{\text{trend},(g)}$ is the prediction after imposing the trend, t_0 is the first year of the window, δ is the imposed cumulative change, and T is the calendar time span of the window. Slopes $\gamma_{s,K}^{(g)}$ were then estimated from the trend-modified predictions using the same centered-year regression described above. Detection was assessed using the same posterior probability criterion as in the observed trend analysis.

Imposed trend detectability answers the question: if a trend of the specified magnitude were to emerge at a site, how many plots would be required to detect it?

3.5.5 Slope estimation stability

The detection-based analyses above evaluate whether an estimated slope exceeds a fixed threshold. A complementary question is whether the slope estimate itself is stable—that is, whether it would yield a similar value if a different set of plots were selected. This is directly relevant to monitoring design: a slope estimate that varies substantially depending on which plots are sampled provides a weak basis for inference, regardless of whether it exceeds a detection threshold.

To evaluate this, we computed the coefficient of variation (CV) of the slope estimate across Monte Carlo replicates at each sample size K . For each species s , window position, and sample size, each of the $R = 50$ Monte Carlo replicates yields a mean slope (averaged across posterior draws):

$$\bar{\gamma}_{s,K,m} = \frac{1}{G} \sum_{g=1}^G \gamma_{s,K,m}^{(g)}$$

where $\gamma_{s,K,m}^{(g)}$ is the slope estimate for species s at sample size K in Monte Carlo replicate m for posterior draw g , and G is the number of posterior draws.

The coefficient of variation across replicates is then:

$$CV_s(K) = \frac{SD(\bar{\gamma}_{s,K,1}, \dots, \bar{\gamma}_{s,K,R})}{|\text{mean}(\bar{\gamma}_{s,K,1}, \dots, \bar{\gamma}_{s,K,R})|}$$

At low K , slope estimates vary widely depending on which plots are selected (high CV); as K increases, estimates converge (low CV). A CV below 0.50 indicates that the standard deviation of the slope estimate is less than half its magnitude, such that variation in plot selection does not qualitatively change the inferred direction and strength of the trend.

This stability analysis provides a perspective complementary to the detection-based analyses. Adding spatially heterogeneous plots to annual site means can increase year-to-year variability, making threshold-based detection more difficult at higher K —an effect that reflects real ecological heterogeneity but complicates interpretation of detection curves. In contrast, the CV metric evaluates the precision of the slope estimate itself and decreases monotonically with K at nearly all sites.

For each species and site, we defined K_{CV50} as the smallest sample size at which $CV_s(K) \leq 0.50$, representing the point at which slope estimates become stable with respect to plot selection.

3.5.6 Definition of N^* for trend detection

For each site, species, and window width, we defined N^* as the smallest number of plots K for which the trend detection criterion was met. As with the counterfactual change detection analysis, N^* values were computed separately for each species and then summarized at the community level.

3.6 Scope and relationship between analyses

The counterfactual year-to-year change detection and trend analyses address ecological change at different temporal scales. The counterfactual year-to-year change detection analysis quantifies the spatial replication required to detect a specified relative change at a single time point, providing guidance relevant to monitoring for abrupt ecological shifts. The trend analysis evaluates the spatial replication required for reliable trend estimation over multi-year windows, providing guidance relevant to tracking gradual directional changes.

These analyses are complementary. The slope stability analysis from the trend framework directly parallels the counterfactual change detection curves: both ask how the quality of inference improves with additional plots. Where both analyses converge on similar sample size requirements, the recommendation is strengthened.

4 RESULTS

We present results from site-level analyses across NEON, drawing on both the counterfactual change detection framework and the multi-year trend analysis. Together, these analyses evaluate how the ability to detect ecological change and estimate directional trends depends on spatial replication (number of plots) under realistic levels of spatial and temporal variability.

The analyses leverage 8–12 years of NEON vegetation data per site, representing a substantial increase in temporal replication relative to the 2018 design optimization effort. This expanded record enables direct evaluation of both year-to-year change detection and multi-year trend estimation within a unified modeling framework.

We begin with a summary of site-level data characteristics, including the number of years of observation and the number of species retained after applying the cover filter. We then present detailed results for a small number of representative sites to illustrate the behavior of each analysis, followed by summary tables that synthesize results across all completed sites.

4.0.1 Site characteristics

Across completed sites, the number of years of observation ranged from 2013 to 2024, with most sites having at least eight years of data. The number of species included in the GJAM model ranged from 39 to 406 per site, reflecting substantial variation in species richness across the Observatory. After applying the 3% mean cover threshold, the number of species retained for analysis was reduced to 4 to 122 per site.

All species were included in model fitting and posterior prediction to preserve community structure. Detection and stability metrics, however, were restricted to species with mean raw cover $\geq 3\%$. Below this threshold, a 20% relative change falls within the resolution of the field protocol and cannot be reliably distinguished from measurement noise. For example, a species with 2% mean cover would require detection of a 0.4 percentage point change under a 20% perturbation, which is smaller than the effective resolution of visual percent cover estimates in the field. This filtering ensures that inference is focused on the portion of the community where meaningful change can be resolved, and the same filtered species set was used consistently across both the change detection and trend analyses.

For each site, we report three species counts: (1) the total number of species observed in the raw dataset, (2) the number of species included in the model, and (3) the number of species retained for inference after applying the 3% mean cover threshold (Table 1).

Table 1: Summary of site characteristics, including years of observation and species counts for observed, modeled, and inference subsets.

Domain	Site	Site Code	Years	Year Range	Species (observed)	Species (modeled)	Species ($\geq 3\%$ cover)
D16	Abby Road	ABBY	8	2017-2024	247	129	43
D18	Utqiagvik	BARR	7	2017-2024	90	56	10
D01	Bartlett Experimental Forest	BART	11	2014-2024	95	44	7
D02	Blandy Experimental Farm	BLAN	10	2015-2024	292	149	30
D19	Caribou-Poker Creeks Research Watershed	BONA	8	2017-2024	84	46	26

Domain	Site	Site Code	Years	Year Range	Species (observed)	Species (modeled)	Species (≥3% cover)
D11	Lyndon B. Johnson National Grassland	CLBJ	7	2016-2024	491	249	63
D10	Central Plains Experimental Range	CPER	12	2013-2024	206	122	22
D09	Dakota Coteau Field Site	DCFS	8	2017-2024	264	143	36
D19	Delta Junction	DEJU	9	2016-2024	182	98	24
D03	Disney Wilderness Preserve	DSNY	10	2014-2024	612	304	92
D07	Great Smoky Mountains National Park	GRSM	5	2015-2024	259	126	47
D04	Guánica Forest	GUAN	10	2015-2024	229	121	77
D01	Harvard Forest & Quabbin Watershed	HARV	11	2014-2024	322	145	47
D19	Healy	HEAL	10	2015-2024	136	78	36
D03	The Jones Center At Ichauway	JERC	10	2014-2024	823	406	122
D14	Jornada Experimental Range	JORN	9	2015-2024	171	91	5
D06	Konza Prairie Agroecosystem	KONA	8	2017-2024	113	46	9
D06	Konza Prairie Biological Station	KONZ	5	2016-2024	355	195	52
D04	Lajas Experimental Station	LAJA	9	2016-2024	226	108	83
D08	Lenoir Landing	LENO	8	2016-2024	289	161	33
D07	Mountain Lake Biological Station	MLBS	8	2017-2024	217	124	48
D13	Moab	MOAB	8	2015-2024	174	95	4
D13	Niwot Ridge	NIWO	10	2015-2024	247	147	30
D09	Northern Great Plains Research Laboratory	NOGP	9	2016-2024	257	125	49

Domain	Site	Site Code	Years	Year Range	Species (observed)	Species (modeled)	Species (≥3% cover)
D11	Marvin Klemme Range Research Station	OAES	9	2015-2024	397	224	47
D15	Onaqui	ONAQ	10	2014-2024	181	90	9
D07	Oak Ridge	ORNL	11	2014-2024	370	186	64
D03	Ordway-Swisher Biological Station	OSBS	11	2013-2024	512	212	54
D20	Pu'u Maka'ala Natural Area Reserve	PUUM	7	2018-2024	122	61	20
D10	Rocky Mountains	RMNP	8	2017-2024	215	98	19
D02	Smithsonian Conservation Biology Institute	SCBI	11	2014-2024	406	226	58
D02	Smithsonian Environmental Research Center	SERC	9	2015-2024	285	171	35
D17	San Joaquin Experimental Range	SJER	8	2017-2024	252	130	40
D17	Soaproot Saddle	SOAP	6	2018-2024	317	131	46
D14	Santa Rita Experimental Range	SRER	9	2016-2024	441	271	38
D05	Steigerwaldt-Chequamegon	STEI	10	2015-2024	418	190	71
D10	North Sterling	STER	11	2014-2024	70	39	16
D08	Talladega National Forest	TALL	11	2014-2024	452	244	64
D17	Lower Teakettle	TEAK	5	2019-2024	199	70	29
D18	Toolik Field Station	TOOL	8	2017-2024	188	105	19
D05	Treehaven	TREE	10	2015-2024	411	191	49
D06	University of Kansas Field Station	UKFS	9	2016-2024	383	207	68
D05	University of Notre Dame Environmental Research Center	UNDE	11	2014-2024	393	198	44

Domain	Site	Site Code	Years	Year Range	Species (observed)	Species (modeled)	Species ($\geq 3\%$ cover)
D09	Chase Lake National Wildlife Refuge	WOOD	11	2014-2024	282	127	45
D16	Wind River Experimental Forest	WREF	7	2018-2024	123	60	23
D12	Yellowstone National Park	YELL	6	2018-2024	421	185	29

4.1 Year-to-year change detection (counterfactual analysis)

4.1.1 Example sites

We illustrate the behavior of the counterfactual change detection analysis using three representative sites: ABBY, GRSM, and BONA (Figure 2), spanning a range of community composition and spatial heterogeneity.

At all three sites, detection increased with sample size at the focal effect size of $\delta = 0.20$, but the rate of increase differed among sites. At ABBY, detection rose steadily with increasing K , approaching high detection probabilities at larger sample sizes. GRSM required a greater number of plots to achieve comparable levels of detection, reflecting increased spatial variability. At BONA, detection increased with K but did not reach the 80% threshold within the evaluated range, indicating that the imposed change could not be reliably detected even at maximum sampling effort.

At smaller effect sizes ($\delta = 0.15$), detection remained low across all sample sizes, indicating that changes below 20% are not reliably detectable under observed spatial variability. At larger effect sizes ($\delta \geq 0.25$), detection increased rapidly and approached saturation at relatively small sample sizes, particularly at ABBY.

4.1.2 Summary across sites

Across sites, the number of plots required to achieve the detection criterion (K^*) varied substantially (Table 2). At $\delta = 0.20$, most sites reached the 80% detection threshold between approximately $K = 20$ and $K = 30$, although several sites did not reach the threshold within the evaluated range.

Detection increased monotonically with sample size at nearly all sites, but the rate of increase differed. Sites with relatively homogeneous spatial structure showed rapid gains in detection with increasing K , whereas more heterogeneous communities exhibited slower, more gradual increases.

Detection at the maximum evaluated sample size ($K = 30$) also varied among sites, indicating that for some communities the 80% detection threshold was not achievable within the range of sampling effort considered. These differences reflect variation in community composition and spatial heterogeneity rather than differences in model behavior.

Species-level detection requirements were substantially lower than the community-level thresholds reported here, indicating that many species are individually detectable at small sample sizes. However, the

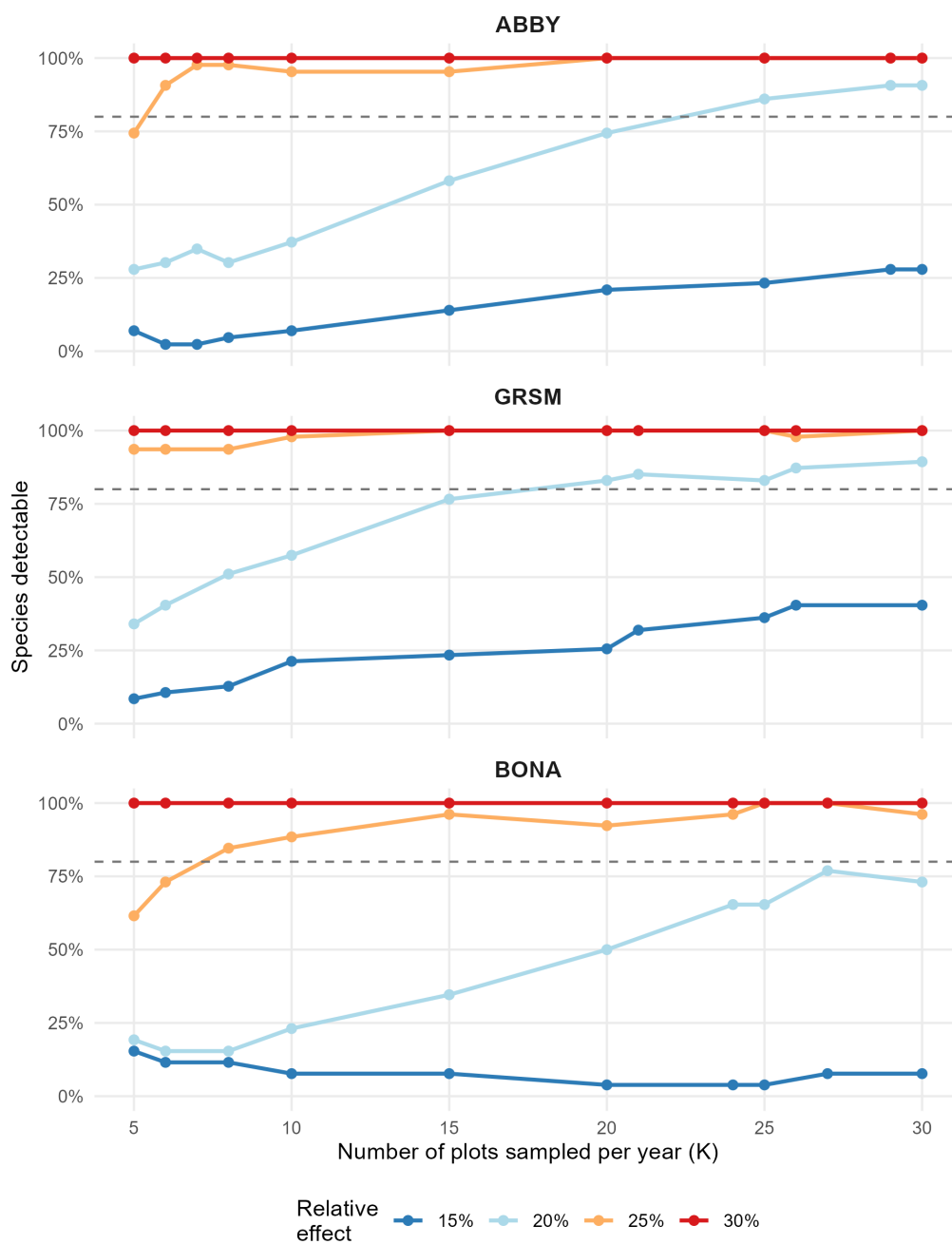


Figure 2: Counterfactual change detection across three example sites. Curves show the percentage of species with detection probability at least 0.80 as a function of the number of plots sampled per year (K), for imposed relative changes of 15–30%.

community-level metric requires that most species exceed the detection criterion, resulting in larger sample size requirements for reliable inference at the community scale.

Table 2: Summary of counterfactual change detection across sites. K^* is the smallest evaluated number of plots at which at least 80% of species exceeded the detection criterion for a 20% relative change. Detection at $K = 30$ reports the percentage of species detectable at the maximum evaluated sample size.

Site	Species ($\geq 3\%$ cover)	K^*	Detection at $K = 30$ (%)
BART	7	5	57.1
JERC	122	10	94.3
SERC	35	10	77.1
SRER	38	10	89.5
TALL	64	10	93.8
CLBJ	63	15	95.2
LENO	33	15	90.9
MOAB	4	15	100.0
SOAP	46	15	91.3
STEI	71	15	88.7
UNDE	44	15	90.9
DSNY	92	20	91.3
GRSM	47	20	89.4
JORN	5	20	80.0
OAES	47	20	87.2
ONAQ	9	20	88.9
ORNL	64	20	92.2
OSBS	54	20	94.4
ABBY	43	25	90.7
HARV	47	25	83.0
KONZ	52	25	88.5
NIWO	30	25	86.7
NOGP	49	25	83.7
WOOD	45	25	86.7
RMNP	19	30	84.2
SCBI	58	30	81.0
SJER	40	30	90.0
UKFS	68	30	82.4
WREF	23	30	82.6

Site	Species ($\geq 3\%$ cover)	K^*	Detection at $K = 30$ (%)
BARR	10	NA	70.0
BLAN	30	NA	NA
BONA	26	NA	73.1
CPER	22	NA	68.2
DCFS	36	NA	69.4
DEJU	24	NA	70.8
GUAN	77	NA	NA
HEAL	36	NA	66.7
KONA	9	NA	66.7
LAJA	83	NA	63.9
MLBS	48	NA	77.1
PUUM	20	NA	NA
STER	16	NA	NA
TEAK	29	NA	NA
TOOL	19	NA	47.4
TREE	49	NA	NA
YELL	29	NA	NA

4.2 Multi-year trend detection

4.2.1 Example sites

The detection curves shown here are based on imposed trend detectability, providing a standardized comparison of sensitivity across sites.

Patterns of trend detectability broadly paralleled those observed for year-to-year change detection, but with important differences arising from the temporal structure of the signal (Figure 3).

At ABBY, trend detection increased steadily with sample size, and both detection and stability were achieved at relatively small K . GRSM required larger sample sizes to achieve comparable levels of detection, reflecting greater variability in annual site means. At BONA, detection increased with K but did not consistently reach the detection threshold, while slope estimates remained sensitive to plot selection even at larger sample sizes. Together, these sites illustrate that strong signals do not necessarily translate to stable inference.

Slope stability showed a consistent and interpretable response to increasing sample size (Figure 4). Across all three sites, the coefficient of variation of slope estimates declined with increasing K , indicating improved precision with additional spatial replication. However, the rate of improvement differed among sites. ABBY achieved low CV values at relatively small K , whereas GRSM and BONA required larger sample sizes to achieve comparable levels of stability.

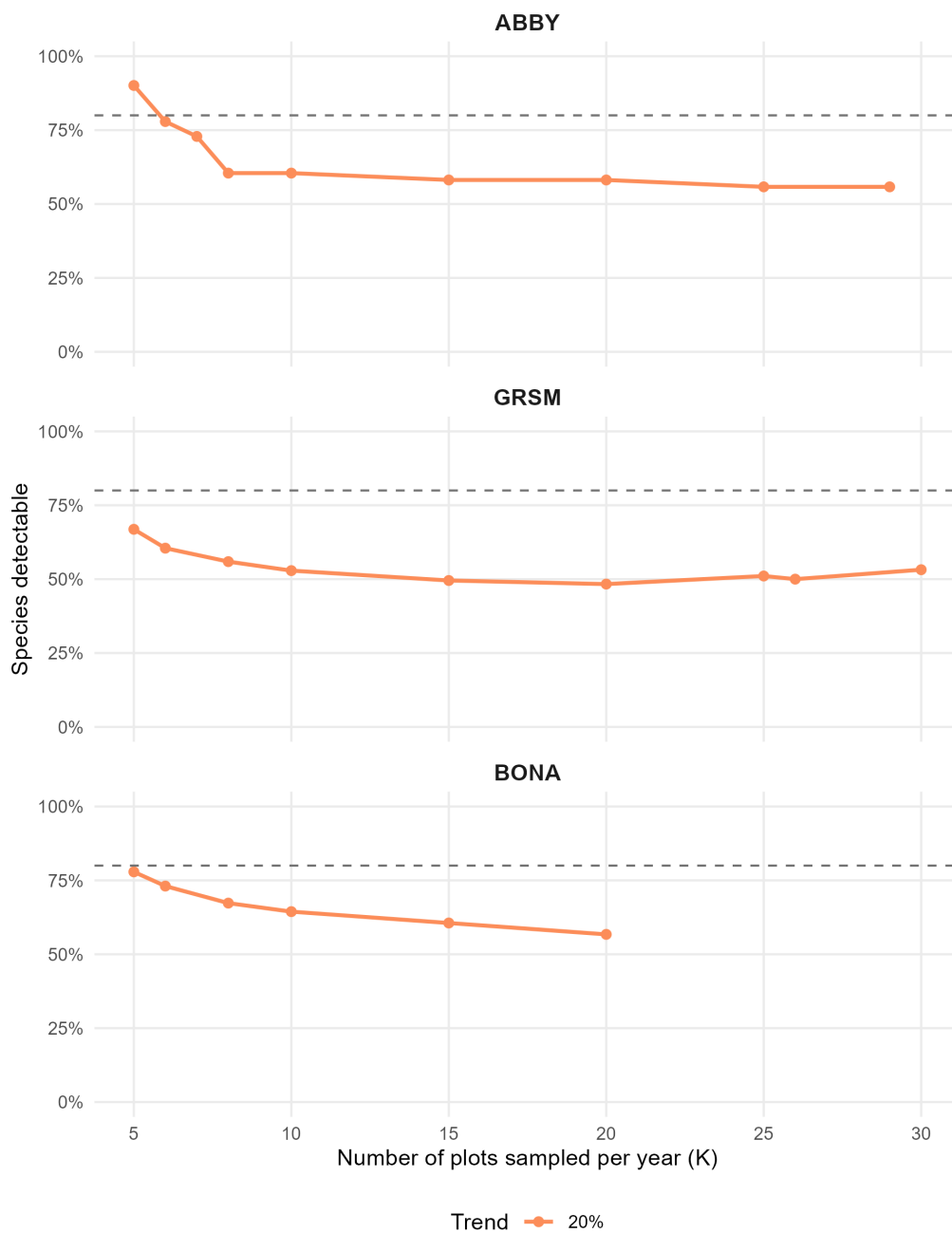


Figure 3: Trend detection across three example sites. Curves show the fraction of species with detection probability at least 0.80 as a function of the number of plots sampled per year (K), for imposed cumulative trends of different magnitudes.

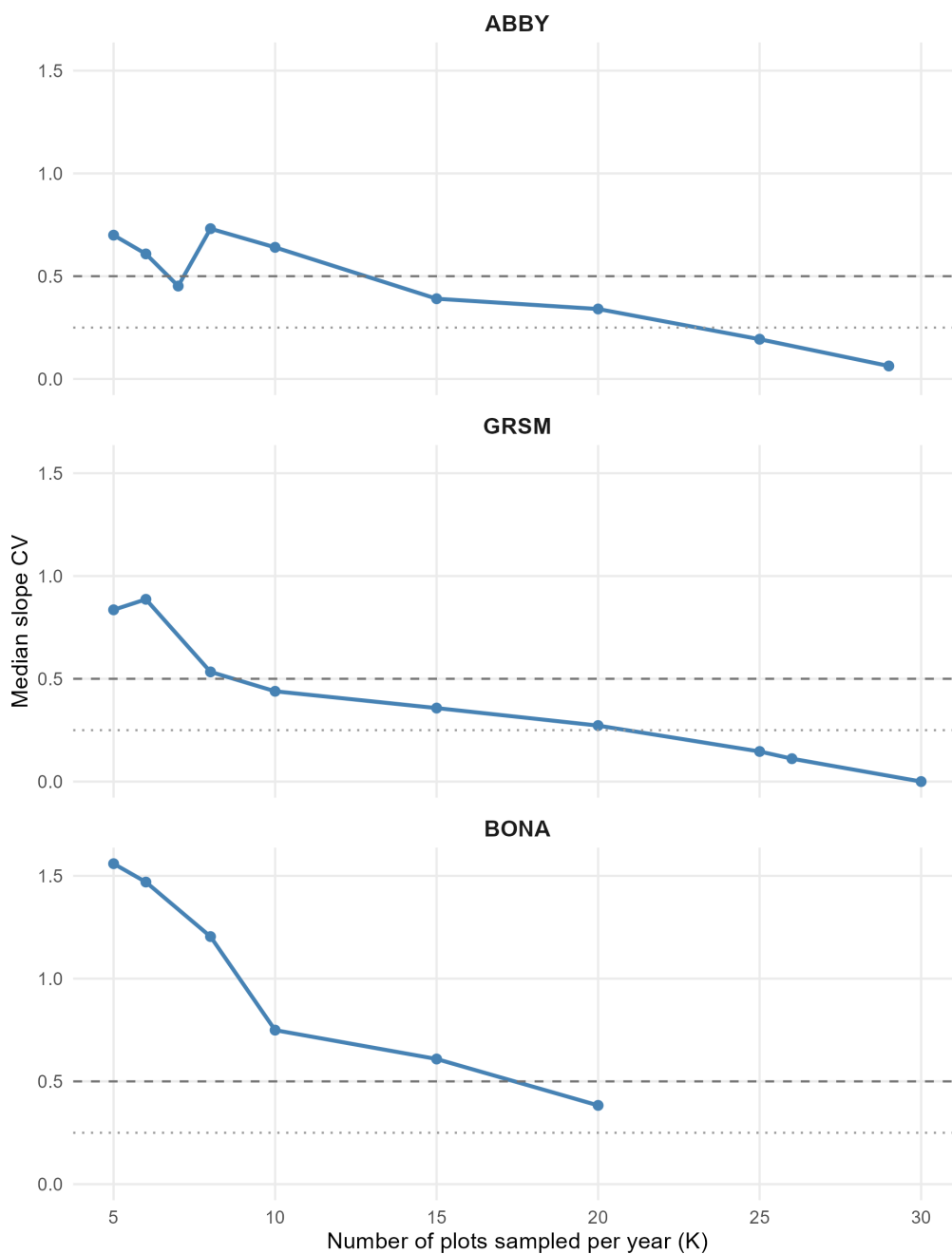


Figure 4: Slope estimation stability across three example sites. Curves show the median coefficient of variation (CV) of species-level slope estimates as a function of the number of plots sampled per year (K). Lower CV indicates greater stability with respect to plot selection. Horizontal lines indicate reference thresholds at CV = 0.50 (used to define stable estimation) and CV = 0.25.

4.2.2 Detection and stability

Detection and stability provided complementary perspectives on trend inference. Detection evaluates whether the estimated slope exceeds a specified threshold, while stability evaluates whether the slope estimate is consistent across repeated subsamples of plots.

Across sites, detection thresholds were often reached at smaller sample sizes than those required for stable estimation, but stability required larger values of K . In many cases, slope estimates remained sensitive to plot selection even after detection thresholds were met, indicating that detectability does not imply reliable estimation of trend magnitude.

The coefficient of variation of slope estimates decreased monotonically with increasing K at nearly all sites, indicating that additional plots consistently improve the precision of trend estimates. In contrast, detection-based metrics can be influenced by spatial heterogeneity in ways that produce non-monotonic or difficult-to-interpret patterns. Stability therefore provides a more direct measure of the reliability of trend estimates under realistic sampling variability.

4.2.3 Summary across sites

Across the Observatory, sample size requirements for trend detection and stability varied substantially among sites (Table 3). In most cases, the number of plots required to detect a 20% cumulative trend (Trend K^*) was smaller than the number required to achieve stable slope estimates (K_{CV50}), indicating that reliable estimation of trend magnitude requires more intensive sampling than detection alone.

At some sites, the detection threshold was not achieved within the evaluated range of sample sizes, even though slope estimates became progressively more stable with increasing K . Conversely, sites with relatively low spatial heterogeneity showed similar sample size requirements for detection and stability.

Together, these results indicate that detectability and stability represent distinct constraints on inference. Detection determines whether change can be identified, whereas stability determines whether that change can be quantified reliably. Both must be considered when evaluating monitoring design.

Table 3: Summary of multi-year trend detection and slope stability across sites. Trend K^* is the smallest evaluated number of plots at which at least 80% of species exceeded the detection criterion for a 20% imposed cumulative trend. Detection at $K = 30$ reports the percentage of species detectable at the maximum evaluated sample size. K_{CV50} is the smallest evaluated number of plots at which the median slope coefficient of variation was less than or equal to 0.50; CV at $K = 30$ reports the median slope CV at the maximum evaluated sample size.

Site Code	Species ($\geq 3\%$ cover)	Trend K^*	Trend detection at $K = 30$ (%)	K_{CV50}	CV at $K = 30$
DCFS	36	5	55.6	5	0
MLBS	48	5	NA	5	NA
SJER	40	5	NA	5	NA
BARR	10	NA	NA	5	NA
CLBJ	63	NA	NA	5	NA

Site Code	Species (≥3% cover)	Trend K*	Trend detection at K = 30 (%)	K_CV50	CV at K = 30
CPER	22	NA	NA	5	NA
GUAN	77	NA	NA	5	NA
JORN	5	NA	60.0	5	0
LENO	33	NA	NA	5	NA
NOGP	49	NA	49.0	5	0
ONAQ	9	NA	NA	5	NA
OSBS	54	NA	NA	5	NA
RMNP	19	NA	68.4	5	0
SOAP	46	NA	NA	5	NA
SRER	38	NA	NA	5	NA
STER	16	NA	NA	5	NA
WREF	23	NA	NA	5	NA
YELL	29	NA	NA	5	NA
MOAB	4	5	NA	6	NA
TREE	49	5	NA	6	NA
STEI	71	NA	NA	6	NA
ABBY	43	5	NA	7	NA
UNDE	44	NA	NA	7	NA
WOOD	45	NA	NA	7	NA
DSNY	92	NA	50.0	8	0
KONA	9	NA	NA	8	NA
OAES	47	NA	62.8	8	0
BLAN	30	5	NA	10	NA
LAJA	83	5	NA	10	NA
BART	7	NA	57.1	10	0
GRSM	47	NA	53.2	10	0
JERC	122	NA	NA	10	NA
TALL	64	NA	NA	10	NA
TEAK	29	NA	NA	10	NA
TOOL	19	NA	NA	10	NA
PUUM	20	NA	NA	12	NA
KONZ	52	NA	NA	13	NA
UKFS	68	5	NA	15	NA
ORNL	64	NA	NA	15	NA
SCBI	58	NA	NA	15	NA

Site Code	Species ($\geq 3\%$ cover)	Trend K^*	Trend detection at $K = 30$ (%)	K_{CV50}	CV at $K = 30$
DEJU	24	5	NA	20	NA
BONA	26	NA	NA	20	NA
HARV	47	NA	38.3	20	0
NIWO	30	NA	50.0	20	0
SERC	35	NA	49.1	20	0
HEAL	36	5	NA	NA	NA

4.3 Synthesis across analyses

4.4 Synthesis across analyses

The counterfactual change detection and trend analyses provide complementary but distinct perspectives on the detectability of ecological change. Both analyses show that increasing the number of sampled plots improves inference, but they differ in what constitutes sufficient sampling.

The counterfactual analysis evaluates the ability to detect a discrete change, and indicates that the detection criterion for a 20% relative change is often met at sample sizes on the order of $K \approx 20$ –25. In contrast, the trend analysis shows that reliable estimation of directional change imposes a more stringent requirement. While trend detection thresholds are frequently reached at similar or smaller sample sizes, the stability analysis demonstrates that larger values of K are required for slope estimates to be robust to spatial subsampling.

Where both approaches converge on similar sample size requirements, detection and estimation are aligned, providing strong guidance for monitoring design. Where they diverge, detection alone can overstate the reliability of inference, as slope estimates may remain sensitive to plot selection even after detection thresholds are met. This distinction reflects the difference between identifying change and quantifying its magnitude.

Together, these results indicate that detectability and stability represent complementary constraints on inference. Detection determines whether change can be identified, whereas stability determines whether that change can be estimated reliably. Evaluating both provides a framework for designing monitoring systems that are robust across multiple temporal scales.

5 DISCUSSION

Monitoring design is constrained not only by how easily ecological change can be detected, but by how it is defined. Across NEON plant communities, we show that sample sizes sufficient for detecting discrete change are often insufficient for reliably estimating multi-year trends.

5.1 Complementary constraints on inference

This analysis shows that conclusions about required spatial replication depend not only on the magnitude of ecological change, but also on how inference is defined. The counterfactual change detection and multi-year trend analyses provide complementary perspectives on this question.

The counterfactual analysis evaluates the ability to detect a specified relative change at a single time point, providing guidance relevant to monitoring abrupt ecological shifts. The trend analysis evaluates the reliability of estimated rates of change over multi-year windows, providing guidance relevant to gradual directional processes.

Across the Observatory, these approaches converge on a consistent range of sampling effort. For a 20% relative change, most sites required approximately 20–25 plots to achieve the community-level detection criterion. A similar range was identified by the trend stability analysis as the point at which slope estimates become robust to plot selection (median $CV \leq 0.50$). The convergence of these independent analyses, which operate on different temporal scales and use different statistical criteria, strengthens confidence in this recommendation.

At the same time, the analyses highlight an important distinction: detection and estimation are not equivalent. Detection thresholds were often reached at smaller sample sizes than those required for stable trend estimation. In many cases, slope estimates remained sensitive to which plots were sampled even after detection thresholds were met. This indicates that detection alone can overstate the reliability of inference, and that stability provides a necessary complement for evaluating monitoring design.

5.2 Stability as a design criterion

The slope stability framework provides a direct and interpretable measure of inference quality. The coefficient of variation of slope estimates quantifies the sensitivity of trend estimates to spatial sampling, and decreases monotonically with increasing sample size at nearly all sites.

The CV threshold ($CV \leq 0.50$) defines an analogue to the K^* concept used in detection-based analyses: it identifies the point at which additional sampling no longer qualitatively changes the inferred trend. This provides a practical and interpretable criterion for monitoring design. Unlike detection thresholds, which depend on an arbitrary choice of effect size, the stability criterion is intrinsic to the data and reflects the spatial heterogeneity of the system.

The contrast between detection and stability is particularly important in the context of trend inference. Detection-based trend metrics can exhibit non-monotonic behavior as additional plots are added, reflecting increased spatial variability in annual site means. This spatial dilution effect is ecologically real, but complicates interpretation of detection curves. In contrast, the stability metric directly evaluates the precision of the slope estimate and provides a consistent basis for comparing sites.

5.3 Per-species scaling and the role of filtering

The use of per-species perturbation and detection thresholds scaled to raw mean cover ensures that all species face equivalent proportional challenges. This avoids the bias inherent in fixed absolute thresholds, where a given percentage-point change has very different ecological meaning across the abundance

spectrum.

The 3% cover floor defines the portion of the community for which meaningful inference is possible. Below this threshold, a 20% relative change falls within the resolution of the field protocol and cannot be reliably distinguished from measurement noise. As a result, the analyses characterize detection and stability for the more abundant component of the community, where ecological change can be resolved with confidence.

At most NEON sites, a large fraction of species fall below this threshold. This does not represent a limitation of the modeling framework, but rather a constraint imposed by the combination of ecological rarity and measurement precision. The filtering step therefore clarifies the scope of inference rather than restricting it.

5.4 Modeling framework and dimension reduction

The use of GJAM enables joint inference across species, accounting for shared environmental responses and residual covariance. This represents a substantial advance over the univariate framework used in the 2018 optimization analysis (Barnett et al. 2018), which could not capture coordinated responses among species.

Dimension reduction (Taylor-Rodriguez et al. 2017) is essential for applying this framework at NEON scales. At most sites, the number of species exceeds the effective sample size required to estimate a full covariance matrix. The reduced-rank representation captures dominant modes of association while maintaining computational tractability, allowing the model to scale across the full diversity gradient of the Observatory.

5.5 Implications for NEON sampling design

The results indicate that NEON's current design of approximately 20–30 plots per site is well-calibrated for detecting 20% relative changes in plant cover and for producing stable trend estimates over multi-year windows. This conclusion is supported by both the perturbation-based detection analysis and the trend stability analysis.

Changes smaller than approximately 15% are not reliably detectable under observed spatial variability at any evaluated sample size, establishing a practical lower bound on sensitivity. Conversely, changes of 25% or greater are detectable at relatively small sample sizes, indicating that large ecological shifts will be captured regardless of sampling intensity.

Site-level variation in required sample size reflects underlying spatial heterogeneity. Homogeneous systems can support reliable inference with relatively few plots, whereas more complex systems require greater spatial replication. The framework presented here provides a quantitative basis for evaluating these differences and adapting sampling strategies accordingly.

5.6 Addressing the 2018 trend detection question

The 2018 design optimization effort identified trend detection as a priority question but could not address it due to limited temporal replication (Barnett et al. 2018). With 8–12 years of data now available, the

moving-window trend analysis presented here directly addresses that gap.

The key result is that reliable trend estimation imposes similar sample size requirements to those identified for detecting discrete changes. The stability framework provides an actionable interpretation of this result, linking sample size directly to the reliability of estimated rates of change.

5.7 Limitations and future directions

Several limitations warrant consideration. First, the analysis is based on a single model fit per site, and does not propagate uncertainty associated with refitting the model under different plot subsets. As a result, the reported sample size requirements represent optimistic estimates of detection and stability. A refit-based analysis would provide a more complete accounting of uncertainty and is a priority for future work.

Second, the analysis focuses on a range of relative effect sizes (15–30%), with 20% representing a critical threshold for detectability. Sensitivity to alternative detection thresholds was not explored and could be evaluated in future work.

Third, the compositional structure of the GJAM model compresses absolute cover values relative to raw field observations. This motivates the use of raw field cover for defining perturbations and detection thresholds, but introduces a layer of abstraction that should be considered when interpreting results.

The framework developed here clarifies that sampling adequacy depends on how ecological change is defined. Sample sizes sufficient for detecting discrete change are not necessarily sufficient for reliably estimating rates of change over time. Across NEON plant communities, both analyses indicate that approximately 20–25 plots are required for robust inference, but for different reasons: detection thresholds are met at this scale, and slope estimates become stable with respect to spatial sampling.

This distinction has direct implications for monitoring design. Designs based solely on detection criteria may underestimate the sampling effort required for reliable ecological inference. Incorporating stability-based metrics provides a more complete basis for evaluating monitoring performance, particularly for questions involving gradual, multi-year change. As NEON and similar observatories continue to mature, distinguishing between detectability and reliability will be essential for ensuring that monitoring design supports not just the detection of change, but its reliable interpretation.

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